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SURVEY

A Mapping and State-of-the-Art Survey on Multi-Objective Optimization Methods for Multi-Agent Systems

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ABSTRACT Over the last decades, researchers have studied the Multi-Objective Optimization (MOO) problem for Multi-Agent Systems (MASs). However, most of them consider the problem formulation to be the sum of objective functions, and no work has reviewed problems with the formulation of the prioritized sum of objective functions to facilitate the study of the subject and identify the needs arising from it. In the context of Multi-Robot Systems (MRSs), most studies only focus on the mathematical development of their proposed MOO algorithm without paying attention to the real application. There is no comprehensive review to identify the reliable algorithms already applied to real platforms. Using a mapping and state-of-the-art review, this paper aims to fill these gaps by first offering a detailed overview of the discrete-time MOO methods for MASs. More specifically, we classify existing MOO methods based on the formulation of the problem into the sum of objective functions and the prioritized sum of objective functions. Secondly, we review the applications of these methods in MRSs and the practical implementation of MOO algorithms on real MRSs. Finally, we suggest future research directions to extend the existing methods to more realistic approaches, including open problems in the new research area of the prioritized sum of objective functions and practical challenges for implementing the existing methods in robotics. This work introduces the field of MAS to researchers and enables them to position themselves in the current research trends.

INDEX TERMS Multi-agent systems, multi-objective optimization, multi-robot systems, prioritized sum of objective functions.

I. INTRODUCTION

MASs, such as a team of robots, can be applied in many emergency scenarios to prevent humans from being harmed. These scenarios include landmine clearance, search and rescue missions, extinguishing indoor and outdoor fires, and exploring unknown environments. Apart from this, they can also be implemented to perform many non-emergency scenarios, including target tracking, collaborative navigation, path-finding, and task assignment.

In other words, a MAS is a set of agents acting in an environment in collaboration to achieve a common goal faster than individually. Sometimes, this common goal can be impossible for an individual agent to reach, such as exploring a vast and complex environment by a single robot, which is impossible considering its limited sensors and battery capacity. Agents work towards fulfilling their tasks by either sharing or not sharing their knowledge with each other. Their coordination may be cooperative, competitive, or both, and the agents may work on the same or different set of goals [1], [2]. Although most MAS applications explore the optimization of agents' policies concerning a single objective, real-world applications are naturally multi-objective [3]. Each of these objectives is presented as an objective function; however, optimizing all of them simultaneously is often challenging since they may

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be contradictory [4]. In other words, improving one of the objectives may negatively impact another. Many MOO techniques have been proposed to solve these problems, including evolutionary algorithms [5] and gradient-based algorithms [6].

MOO has been studied for a long time. Still, recently, it has attracted significant interest [7] due to its potential to solve real-world problems and its various applications in robotics [8], machine learning [9], power systems [10], wireless networks [11], and many more. In a multi-agent context, MOO techniques study conflicting objective functions on a region called the Pareto front to compute the Pareto-optimal solutions [12].

In robotics problems, the usage of MOO is limited to a few applications, including task assignment [13], pathfinding [14], formation control [15], and simultaneous localization and mapping (SLAM) [16]. In the majority of these applications, MOO methods have only been applied to single-agent systems. Furthermore, optimization problems arising in robotics problems are usually non-convex, while most of the existing MOO methods designed for robotics applications consider the problem to be convex. As a result, they do not meet the challenges of MOO applied to MRS. Another reason for having limited applications of MOO algorithms in MRS is that these algorithms often do not consider communication limitations. For large groups of agents using vision techniques, for example, the observed parameters are of large dimensions. This incurs high communication costs, while there may be limited communication capabilities.

Thus, a structured review of the existing methodologies and their characteristics is necessary to identify the gaps in the subject and ensure that research efforts will meet the needs of the robotics field. Therefore, in this paper, we review the literature on MOO methods for MASs along with their applications in MRS. This review provides valuable background for researchers desiring to take their first step in exploring this research area. Moreover, we provide a review of the research on MOO methods implemented on real-time MRS, identifying reliable algorithms and highlighting further the current limitations and gaps in the real-time implementation of MOO methods on MRSs. These gaps include considering delays in the system and insufficient communication infrastructure, which are often omitted in many MOO algorithms.

The classification of MOO methods, in our paper, is based on the formulation of the problem, i.e., the sum of objective functions and the prioritized sum of objective functions. This taxonomy allows us to present a structured view of the existing methods and identify promising future research directions. As a result, we pinpoint the fact that problems with the formulation of the prioritized sum of objective functions have not been deeply investigated in the literature, and there is no work implementing them on real-time platforms to validate their applicability. To the best of our knowledge, this is the first paper reviewing the MOO problems with the formulation of the prioritized sum of objective functions.

This paper mainly contains the following contributions:

1) It provides a detailed and comprehensive survey of the recent MOO methods applied to MASs, along with their applications.
2) It classifies the existing MOO algorithms in terms of formulating the problem into the sum of objective functions and the prioritized sum of objective functions. The first category is further classified into different classes based on the challenges addressed in the literature for that class. This is the first work to review problems with the formulation of the sum of objective functions, providing the researchers with important gaps in this new line of research.
3) It discusses the applications of MOO problems to MRS. Although many applications can be imagined for MOO in MRS, few have been implemented in the literature. We review the implemented applications.
4) This is the first review paper that surveys MOO methods implemented on real MRS. This review will identify safe and reliable algorithms designed so far. Ensuring the applicability and safety of algorithms is very important since implementing unreliable algorithms could have serious consequences. In the human-robot interaction application, for example, failing to perform a task may result in loss of life.
5) It proposes future research directions in MOO problems for MASs and MRS to identify and highlight the areas that need more attention from researchers.

There are also other ways of classifying the MOO methods, as we discuss in Sect. II. However, classifications are not complete, and overlapping may occur. In other words, some methods may fall into more than one class or there may be combinations of different classes. In our paper, we chose to classify the problems based on the formulation since this classification simplifies the understanding of different MOO methods and their differences. That is because methods falling into the same classes often show similar characteristics. We also identify future research directions. Therefore, our review is a mapping review on the subject. On the other hand, we mainly aim to address current approaches on the subject and highlight an area that is in need of further research. That is to say, another part of our study falls within a state-of-the-art review. This new research area is solving the MOO problem with the formulation of the prioritized sum of objective functions. Our review is the first comprehensive review paper to identify the gaps and future research directions for this problem. This helps draw attention to a research subject that researchers omitted while having important advantages, such as improving agents' decision-making power.

Note that we only review first-order, discrete-time decentralized optimization algorithms. In the next paragraphs, a brief description of centralized and decentralized optimization algorithms along with the reason behind our selection is given.

A. CENTRALIZED OPTIMIZATION METHOD

In this optimization method, there is a central node that has access to the entire agents' information. In each iteration of the optimization process, all agents' information is sent to this central node. The objective functions are predefined and fixed, and the central node computes the overall objective function based on the information it has received from agents. Then, it sends the result back to agents, notifying all or some of them of the optimal selection. Thus, no local-to-local communication between agents exists, and agents cannot make decisions individually. High communication is required in this method as all the agents need to be in contact with the central node. As a disadvantage, if the central node gets out of reach or the communication lines fail, the whole system may fail. Furthermore, the computational burden is considerable [17].

B. DECENTRALIZED OPTIMIZATION METHOD

Decentralized optimization strategies have been proposed to overcome the limitations of centralized methods. In these approaches, the agents locally exchange information with each other to mutually improve their knowledge of the environment, and the decision-making power is distributed across all agents. Therefore, the agents can decide individually based on their previous information and the previous information of their neighbors. An example is an estimation scenario for a group of robots that need to localize a target given sensor measurements. They can compute the optimal solution individually, given only their own observations (centralized method), or act in a networked system of robots. In the latter case, they compute the optimal solution given all robots' observations (decentralized method) [18]. Generally, agents may have different priorities in this approach, like travel time, stop time, or fuel consumption in a fleet of self-driving cars [6].

The majority of recent MOO algorithms fall into the second category because of the advantages they offer over the first one. For instance, the convergence rate of decentralized methods may be faster, enhancing the efficiency of the fleet [19]. Moreover, the autonomy of the agents may make the fleet more flexible and more robust compared to centralized methods [20]. A small change in the network can, for example, necessitate a redesign of the centralized algorithm [21]. More importantly, recent advances in technology and having easy access to computing units, such as mobile phones or sensors equipped with a central processing unit, have put increasing attention on decentralized optimization methods [22].

In particular, our paper aims to survey papers published recently (we survey publications after 2000, but we mainly focus on the most recent ones). Thus, only decentralized optimization algorithms are included in the paper.

The rest of the paper is organized as follows: Sect. II presents the existing survey papers on the subject. Sect. III provides some background on graph theory, MOO problems, and Pareto optimality. Problems with the formulation of the sum of objective functions and the prioritized sum of objective functions are reviewed in Sect. IV and Sect. V, respectively. Sect. VI reviews applications of MOO in MRSs and publications with the practical implementation of MOO on MRSs. We propose some future research directions in Sect. VII, and finally, Sect. VIII concludes the paper.

II. EXISTING SURVEYS

In this section, we will mention some review papers on MOO, MASs, and MRSs. The aim is first to explain the difference between our survey and the existing ones, which clarifies the necessity of our paper. Secondly, we will mention some other survey papers that can provide readers with a background or other useful materials on the topic.

MOO techniques have been studied since the 1970s, and several comprehensive review papers on MOO methods have been published since the 1980s. There are many papers reviewing MOO methods for specific applications, such as wireless sensor networks, chemical process engineering, energy forecasting, and so forth [23], [24], [25], [26], [27]. A more general work is [12], in which authors first conduct a survey on how to model multiple objectives to solve MOO problems. Then, they categorize the papers published between 2000 and 2016 into three classes based on the characteristics of objectives, i.e., whether the work deals with system objectives for global welfare and/or individual objectives for individual welfare. The publication [17] is another comprehensive review paper that classifies the distributed MOO algorithms, which are a subcategory of decentralized MOO methods, into discrete-time and continuous-time algorithms. The focus of this paper is on the applications of distributed MOO algorithms to the coordination of distributed energy resources. More recently, a review paper presented in [28] classified distributed MOO algorithms based on the constraints of the problem.

Considering review papers on MASs, [29] presents an overview that enables readers to gain a thorough knowledge of MASs along with their applications and challenges. Other useful background material on MASs, their control and coordination schemes can be found in [30], [31], [32], and [33].

For those interested in applying different MOO algorithms on real robots, [34] presents a comprehensive survey on different platforms used in swarm robotics. This paper mainly explores the reason behind the lack of exploitation of swarm robotic systems in real-world applications. The publications [35], [36] are other review papers discussing different research platforms and hardware architecture to investigate the application of swarm algorithms to robots.

Among the papers surveying MOO methods, [18] is the only work that specifically explores the applications of distributed optimization methods (not decentralized optimization methods) to MRS. This publication categorizes

the distributed optimization algorithms into three classes, named distributed gradient descent, distributed sequential convex programming, and alternating direction method of multipliers (ADMM). Then, it compares the performance of these algorithms for applications in robotics. Our survey differs from [18], in the sense that it studies all the MOO methods applied to MASs, and not only distributed optimization methods. In other words, we survey problems with both the sum of objective functions and the prioritized sum of objective function formulation. The classification, in our paper, also differs from that in [18]. Additionally, we review papers with a practical experiment on real robots, that is not included in [18].

Thus, our paper is the first review of the MOO problems with the problem formulation of the prioritized sum of objective functions and the first review of the practical implementation of MOO algorithms on real-time robots. Although the number of publications conducting research on the mentioned areas is low, having a detailed review of the topic is essential to identify the gaps and highlight the need for more research on the topic.

III. PRELIMINARIES

In this section, we introduce some background on graph theory, MOO, and Pareto front (see [37], [38], [39]) to explain the problem and the terms we use to make the paper more understandable for readers.

A. GRAPH THEORY

MASs and their interactions are mostly modeled using graph theory. More specifically, nodes in a graph represent the agents, and edges represent communications and interactions. In this section, we review basic graph theory definitions. More details can be found in [37].

Let $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ be a graph of order n where $\mathcal{V} = \{1, \ldots, n\}$ is the set of nodes and $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ is the set of edges. A directed edge between node i to node j is denoted by $(i, j) \in \mathcal{E}$, meaning that nodes i and j are adjacent or that agent i is a neighbor of agent j and communicates with it. The set of neighbors of node i is denoted by $N_i = \{j \in \mathcal{V} : (j, i) \in \mathcal{E}\}$. The number of agents with whom agent i communicates is the degree of agent i, denoted by $deg(i)$. A graph is undirected if and only if $(i, j) \in \mathcal{E}$ implies $(j, i) \in \mathcal{E}$. For simplification purposes, it is assumed that the graph does not have any self-loop for all $i \in \mathcal{V}$, i.e., $(i, i) \notin \mathcal{E}$ although all agents have access to their own information. An undirected graph $\mathcal{G}$ is *fully connected* or *complete* if there is an edge between every pair of nodes [37]. A directed path is a sequence of ordered edges in the form of $(i_1, i_2), (i_2, i_3), \ldots$, where $i_j \in \mathcal{V}$. The node j is reachable from node i if a directed path exists from node i to node j. A directed graph $\mathcal{G}$ is *strongly connected* if there is a directed path from every node to every other node [37]. For directed graphs, the out-degree of node i is the number of other nodes to whom agent i sends information. Let $B(\mathcal{G}) = [b_j^i] \in \mathbb{R}^{n\times n}$ be the adjacency matrix associated with the graph $\mathcal{G}$, where b_j^i is the entry in the jth row and ith column, defined as:

$$b_j^i = \begin{cases} 1 & \text{if } (i, j) \in \mathcal{E} \\ 0 & \text{otherwise} \end{cases}$$

B. MULTI-OBJECTIVE OPTIMIZATION

The problem of MOO, which was originally proposed in [40], is defined as follows:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbf{A}} \quad & \mathbf{f}(\mathbf{x}) = \sum_{i=1}^{n} f_i(\mathbf{x}), \\ \text{s.t. } & \mathbf{x} \in \mathbf{X}, \\ & \mathbf{g}(\mathbf{x}) \leq 0, \\ & \mathbf{h}(\mathbf{x}) = 0, \end{aligned} \tag{1}$$

$\mathbf{x} \in \mathbb{R}^n$ is a vector of n decision variables, where n represents the number of agents. $\mathbf{f}(\mathbf{x})$ is the global objective function. f_i, for all $i \in \{1, 2, \ldots, n\}$, is the local objective function known only to agent i. $\mathbf{X} = \{X_1, X_2, \ldots, X_n\}$ is a vector of the local constraint set. $\mathbf{g}(\mathbf{x}) \in \mathbb{R}^m$ and $\mathbf{h}(\mathbf{x}) \in \mathbb{R}^p$ are vectors of global inequality and equality constraint functions, respectively, imposed on decision variables, where m and p are the numbers of global inequality and equality constraint functions, respectively. $\mathbf{A} \in \mathbb{R}^n$ represents the feasible region determined by the constraints, and any point $\mathbf{x} \in \mathbf{A}$ gives a feasible solution.

Note that not all algorithms proposed to solve (1) consider the problem to have all the local and global constraints. Some may only have local constraints, some may only have equality or inequality global constraints, and others may have both local and global constraints.

The weight matrix of a MOO problem, W, is defined as an $n \times n$ matrix, with the scalars $a_j^i(k)$ being the elements of the matrix ($i, j \in \{1, 2, \ldots, n\}$), while $a_j^i(k)$ is the weight that agent i assigns to the information received from a neighboring agent j at the time k.

W is non-negative if all of its entries (a_j^i) are non-negative. W is positive if all of its entries (a_j^i) are positive. A non-negative matrix W is stochastic or column stochastic if the entries of each column sum to 1. A non-negative matrix W is row stochastic if the entries of each row sum to 1. A non-negative matrix W is doubly stochastic if the entries of each row and column sum to 1.

The aim of the MOO problem is to minimize all the objective functions simultaneously. However, the objective functions are usually in conflict with each other, i.e., an improvement in one of them may result in a lower fitness in others. So the problem is to minimize different objective functions in the best way possible [39].

C. PARETO FRONT

Solving a MOO problem is generally associated with building a Pareto front. In this section, some basic information and the definition of Pareto optimality and Pareto front are discussed.

After solving the MOO problem, many feasible solutions are found. However, only a small subset of these solutions

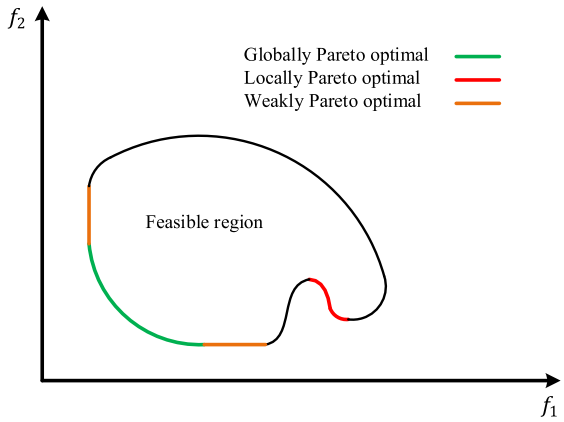

FIGURE 1. Optimality in the Pareto sense [41], [42].

is interesting. This subset only includes solutions that have a domination relation with other ones. In other words, none of them can be improved without the deterioration of at least one other solution.

Mathematically, we say a solution $\mathbf{x}_1 \in \mathbf{A}$ dominates another solution $\mathbf{x}_2 \in \mathbf{A}$, if $f_i(\mathbf{x}_1) \leq f_i(\mathbf{x}_2)$ for all $i = 1, 2, \ldots, n$ and $f_i(\mathbf{x}_1) < f_i(\mathbf{x}_2)$ for at least one i.

The set of all Pareto optimal solutions is called the Pareto front. Fig. 1 illustrates the definition of globally, locally, and weakly Pareto optimal. This figure aims to make the definition of optimality in Pareto sense more straightforward.

A solution $\mathbf{x}^*$ is called (globally) Pareto optimal or (strongly) non-dominated, if and only if there exists no solution $\mathbf{x} \in \mathbf{A}$ such that $f_i(\mathbf{x}) \leq f_i(\mathbf{x}^*)$ for all $i = 1, 2, \ldots, n$ and $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for at least one i.

A solution $\mathbf{x}^*$ is called locally Pareto optimal, if and only if there exists $\delta > 0$ such that there is no solution $\mathbf{x}$ which dominates the solution $\mathbf{x}^*$ with $\mathbf{x} \in \mathbf{A} \cap \mathbf{B}(\mathbf{x}^*, \delta)$, where $\mathbf{B}(\mathbf{x}^*, \delta)$ represents a bowl of center $\mathbf{x}^* \in \mathbf{A}$ and of radius δ, i.e., $\mathbf{B}(\mathbf{x}^*, \delta) = \{\mathbf{x} \in \mathbb{R}^n \mid \| (\mathbf{x} - \mathbf{x}^*) \| < \delta\}$ [39]. If $\mathbf{x}$ is globally Pareto optimal, it is also locally Pareto optimal; however, the reverse does not always hold.

A solution $\mathbf{x}^*$ is called weakly Pareto optimal, if and only if there exists no solution $\mathbf{x} \in \mathbf{A}$ such that $f_i(\mathbf{x}) < f_i(\mathbf{x}^*)$ for all $i = 1, 2, \ldots, n$.

IV. PROBLEM FORMULATION OF SUM OF OBJECTIVE FUNCTIONS

In this section, an overview of some existing decentralized MOO methods along with different extensions of these methods is provided. We classify the algorithms on the basis of the reformulation of the problem, i.e., do they solve the original problem with respect to primal variables, or do they reformulate it as the Lagrangian function to be solved with respect to either both primal and dual variables or only dual variables. The reason behind choosing this classification is that the methods falling into the same classes usually show similar characteristics. This feature, along with having the same formulation, makes it more understandable for readers. Subsequently, this results in a better choice of method for researchers who want to explore the subject.

A. PRIMAL METHODS

In this section, we review the most known primal methods for solving the MOO problem.

1) DISTRIBUTED SUBGRADIENT (DSG) METHOD

DSG algorithm minimizes the problem by averaging the local gradient descent computed by each agent. The algorithm is capable of converging to the optimal solution on non-differentiable convex functions with subgradients. Moreover, there are many extensions of this type of algorithm, offering fast convergence rates. These characteristics have made it ideally suitable for a wide range of applications, such as resource allocation [43] and motion planning [44]. However, works presenting this method mostly develop the math behind the algorithm without application or simulation results.

The first work introducing the DSG algorithm and analyzing its convergence for problems with locally convex and potentially non-smooth objective functions is [43]. In the proposed algorithm, agents estimate the solution and communicate it to other agents asynchronously and over a time-varying network. More specifically, in each iteration of the algorithm, agents update their estimates based on the information received from their immediate neighbors and the subgradient information of their own cost function. In other words, a combination of a local subgradient step and a consensus step (weighted average with immediate neighbors' variables) is done in every iteration of the algorithm.

The framework the authors of [43] used for the analysis of distributed computation models was developed in [40]. This is also the case for most of the other publications developing the DSG algorithm. So, they first introduce a transition matrix and analyze its convergence behavior. Then, they use the convergence rate of the transition matrix in the analysis of the subgradient method.

A constant step-size is used in this algorithm because the authors want it to converge to the optimal solution fast. This causes a constant error term in the estimate of the solution that is proportional to the step-size value. Thus, with a fixed step-size, the DSG algorithm only converges to a neighborhood of the optimal solution [45]. Using diminishing step-size in the algorithm is also possible, but slows down the convergence rate.

Different extensions of DSG methods have been studied in the literature, including locally constrained problems and problems with time-varying graphs or uncoordinated step sizes. In the next paragraphs, we will discuss these extensions with an overview given in Fig. 2.

For diminishing step-sizes, [44] is the first work to develop the DSG algorithm for locally constrained MOO problems and over time-varying networks. Each agent in this algorithm

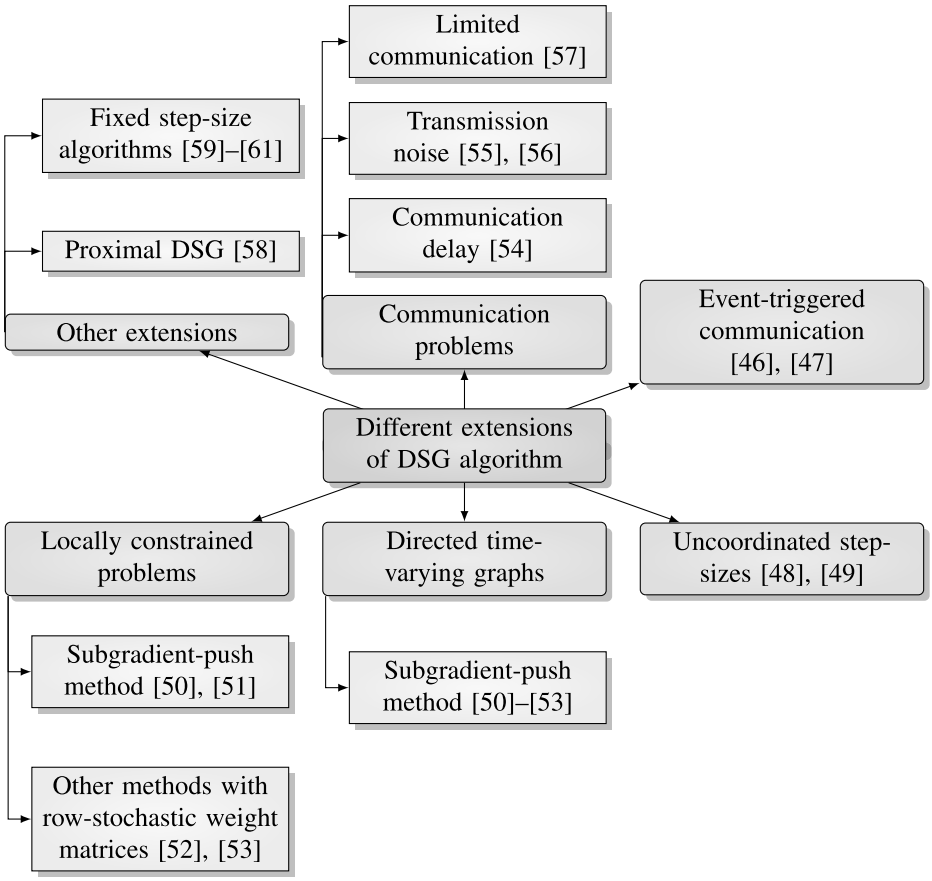

FIGURE 2. Extensions of DSG methods.

performs the DSG algorithm, then takes the projection of their estimate on its individual constraint set. Under the assumptions of the doubly stochastic weight matrix and strongly connected graph, and with an appropriately selected step size (converging to zero fast enough), the authors proved that the algorithm converges to the optimal solution in both cases of having identical or non-identical local constraints. Other DSG algorithms designed for locally constrained MOO problems are presented in [47], [52], [55], [62], and [63]. In [47], [52], and [63], the assumption of having doubly-stochastic weight matrices have been reduced to only row-stochastic matrices. Doubly-stochasticity is usually assumed in many works since it simplifies the proofs of convergence at the cost of restricting communication among agents [6].

Overall, the majority of existing constrained DSG algorithms are designed for locally constrained MOO problems without equality or inequality global constraints. However, there are some publications that combined DSG methods with other methods to present a solution to the globally constraint MOO problem. For problems with global inequality constraints, for instance, the publication [64] proposed a combination of a stochastic subgradient method with a dynamic consensus tracking method for convex, non-smooth cost functions, and non-affine constraint functions. Another work developing a primal-dual (PD) subgradient method for problems with global equality and inequality constraints is [65] (PD methods will be reviewed in the next subsection).

For directed time-varying graphs, most existing algorithms are based on the push-sum method. It is a protocol for node interaction that allows nodes to compute aggregates and average in the graph with directed communication links [50].[1] The reason for using this method is that not all directed graphs accept doubly stochastic weight matrices. On the other hand, the push-sum method reduces the assumption of having doubly stochastic weight matrices to only column-stochastic ones. Authors of [50] were the first to propose a subgradient protocol for directed time-varying graphs, termed the subgradient-push. The proposed algorithm does not require any knowledge of the graph sequence or the number of agents but every agent has to know its out-degree[2] at all times. Moreover, the cost functions need to be convex (not strongly convex), and the algorithm converges at

[1]The convergence of this protocol is illustrated in [66], [67], and [68].

[2]The out-degree of an agent i is the number of agents to them agent i sends information.

a rate of $O(\ln t/\sqrt{t})$, where t is the number of iterations of the algorithm. The same authors developed the subgradient-push method for strongly convex cost function too [51]. With this stronger assumption, they proved the convergence of their algorithms with a rate of $O(\ln t/t)$ even if agents' information about their own cost function is noisy. In both [50] and [51], the weight matrices need to be column stochastic, which necessitates each agent to know its out-degree. To eliminate this requirement, some algorithms have been proposed in the literature that only require row-stochastic weight matrices. One such algorithm is proposed in [52] for convex and Lipschitz continuous cost functions over time-varying general unbalanced directed graphs. Under the requirement of having uniformly jointly strongly connected directed graphs and row-stochastic weight matrices, the authors proved that the algorithm asymptotically converges to the exact optimal solution. Another algorithm is proposed in [53], that also relaxes the compactness assumption of the constraint sets.[3] The authors removed the assumption of boundedness of constraint sets and replaced it with a milder assumption, called the regularity assumption, which can be applied to problems with non-identical and unbounded constraint sets.

Although most of the existing DSG algorithms have been developed for coordinated step-sizes, i.e., when the step-size is the same for all agents in the network, there are some papers investigating uncoordinated step-size [48], [49]. The uncoordinated step-size is practical in cases where the communication is restricted or the communication links are not always reliable. When having an uncoordinated step-size, each agent can update its estimates independently of other agents.

Other extensions of DSG are also available in the literature, such as Proximal DSG [58], but apart from all the extensions mentioned so far, there are some other completely different extensions of DSG algorithms that use fixed step-size in their algorithms and can achieve fast convergence to the exact optimal solution. In other words, they converge much faster than DSG algorithms, with the same computational complexity [69]. These extensions include the DIGing algorithm [59], EXact firsT-ordeR Algorithm (EXTRA) [60], and Network InDependent Step-size algorithm (NIDS) [61]. Indeed, these methods can be reinterpreted as PD methods (which will be investigated in detail in Subsect. IV-B) while not explicitly introducing dual variables. For instance, see [70] and [71] to find the interpretation of EXTRA as a PD method.

Overall, primal methods are effective and easy to implement; however, they use decaying step size in their algorithm, resulting in a slow rate of convergence. With a fixed step size, they only converge to a neighborhood of the optimal solution [59]. The communication burden is also more in primal algorithms with a decaying step size. Moreover, these methods are designed for non-constrained or locally constrained MOO problems, meaning they do not handle problems with global equality or inequality constraints. Even algorithms designed for non-constrained or locally constrained MOO problems focus mainly on formal proofs of performance rather than applications, experiments, or simulations. To address these issues, many methods have been proposed that use fixed step size, including PD methods. PD methods use Lagrangian functions and work with them instead of the original problem. This way, the constrained problem is replaced with a non-constrained problem capable of converging to the exact optimal solution while enjoying the benefits of a fixed step-size algorithm. Apart from that, the applicability of PD methods in real-time has been proved in the literature, as we will discuss in Sect. VI.

B. PD METHODS

In the PD method, which was first introduced by Uzawa [72], the constrained problem of (1) with only convex global inequality constraint functions, $\mathbf{g}(\mathbf{x})$, and strictly convex $\mathbf{f}(\mathbf{x})$ reduces to finding the saddle point of the following Lagrangian, based on the Kuhn-Tucker Theorem [73]:

$$L(\mathbf{x}, \mu) = \mathbf{f}(\mathbf{x}) + \mu^T \mathbf{g}(\mathbf{x}) \tag{2}$$

with Kuhn-Tucker multipliers $\mu_j \geq 0$ for all $j = 1, \ldots, m$. By introducing these multipliers, an equivalent unconstrained problem can be solved instead of (1).

Different papers have proposed different algorithms to find the saddle point of (2). In [74], Uzawa's algorithm [72] is used, which is the first cloud-based optimization algorithm presented for this purpose. The authors named it a quasi-decentralized approach because the computation in this method is distributed between the cloud and the agents. However, It can still be considered a centralized method as the algorithm cannot be executed if the cloud is missing. Moreover, the agents do not have any communication with each other in the proposed architecture. Another work in this direction, which assumes local communication between agents in the presence of the central cloud and mild smoothness conditions imposed on the objective and constraints functions,[4] is [75]. Here, communication delays with the cloud have been accounted for. In this work, the problem of finding the saddle point of (2) is restated as a variational inequality. Then, a fixed Tikhonov regularization [76] with respect to both primal and dual variables is used to regularize the Lagrangian function. The authors present an algorithm that guarantees the closeness of the solution to the saddle point of the unregularized Lagrangian function if the regularization parameters are small enough. Another asynchronous algorithm that uses a Tikhonov regularized primal-dual method and considers communication delays without a uniform bound is [77], in which, the convergence rate in both the primal and the dual spaces is provided.

[3] A subset $S \subset R$ is compact if S is closed and bounded.

[4] It is assumed that objective and constraints functions are convex and continuously differentiable.

In all of these three algorithms, the cloud is responsible for updating the dual variables and the agents are responsible for updating the system's primal variables. In the last two ones, an inseparable global cost is imposed on agents apart from the local costs $\mathbf{f}(\mathbf{x})$, which depends on the states of all agents. Assuming this inseparability in constraints appears more practical in general cases as decision variables are sometimes strongly connected, and no other algorithm can work effectively in this case.

Some problems, such as the Credit Assignment Problem [78], may be solved more efficiently by centralized methods [75], and that is the reason for using a hybrid architecture (combination of the centralized and the decentralized methods) in the literature. However, these algorithms are only useful when having access to a central computer with large processing power that can communicate with all the agents. Otherwise, there might be a time cost.

Other PD algorithms without centralized dual updates are also available in the literature [79], [80], [81], [82], and [83]. In particular, in [81], a PD algorithm is presented under the assumption of a strongly-concave Lagrangian function, which is asynchronous in primal communication and updates and in distributed dual updates. This algorithm is designed for minimizing the single-objective optimization problem, but as the authors assume that the constraint set, $\mathbf{X}$, can be decomposed into $\mathbf{X} = \mathbf{X_1} \times \ldots \times \mathbf{X_n}$, their algorithm can also be applied for a MAS. In this publication, the Lagrangian function is regularized only with respect to μ, and instead of regularization with respect to $\mathbf{x}$, the authors imposed the assumption of diagonal dominance on the Hessian matrix of the Lagrangian function. This is the first publication developing asynchronous PD variable updates for constrained, convex problems of a general form. However, the regularization for convergence guarantee causes unavoidable but bounded convergence errors. Thus, developing algorithms whose convergence does not depend upon system parameters or regularization is an open research question in this direction and can eliminate convergence errors.

As stated earlier, the aim of the reformulation of the problem using the Lagrangian function is to solve it through a fixed step-size algorithm, which can converge to the exact optimal solution. However, there exist some convergence errors in the algorithms proposed by references mentioned so far. This issue has been addressed by proposing an augmented Lagrangian (AL)-based PD method and ADMM.

1) AL-BASED PD METHOD

AL-based PD method, also known as the augmented Lagrangian method (ALM), is a PD method applicable to problems with equality constraints. In this method, the reformulation of (1) can be done through an AL function formed by adding a penalty parameter to the Lagrangian function:

$$L(\mathbf{x}, \mu) = \mathbf{f}(\mathbf{x}) + \frac{\rho}{2}||\mathbf{g}(\mathbf{x})||^2 + \mu^T \mathbf{g}(\mathbf{x}) \tag{3}$$

where μ is the dual variable, and $\rho \geq 0$ is the AL penalty parameter. By (3), the task of solving (1) is converted into solving a sequence of easier unconstrained problems with respect to $\mathbf{x}$, where the influence of the constraints is controlled through the dual variable, μ.

ALM was first introduced in [84], but it has gained increasing attention recently because it can be applied to large-scale problems and because of its flexibility towards how primal problems can be solved. In other words, there are a lot of different methods to perform the primal and dual updates [71]. Some examples are as follows, with an overview presented in Table 1:

- In a multi-step dual accelerated (MSDA) method, first introduced in [85] for smooth and strongly-convex objective functions, exact minimization of Lagrangian function is used to update the primal variables, and Chebyshev acceleration is used to update dual variables. This work was then developed in [86] and [87] for other settings, such as non-smooth objective functions. However, these methods may be inefficient in practice, because they require exact minimization at every iteration, which incurs high computational cost [89].
- Some works [82], [88], [91] proposed inexact minimization of primal and dual variables. In [88], for instance, an inexact Nesterov gradient method is used to update the primal variables. The computational complexity of these inexact methods is studied in [92].
- In a Flexible Primal-Dual framework (abbreviated as FlexPD), developed in [89], an arbitrary finite number of gradient descent steps is done to update the primal variable, without presenting a theoretical analysis of the optimal choice of the number of steps, and a gradient ascent step to update the dual variable. The linear convergence of the algorithm to the exact optimal solution is established for strongly convex and Lipschitz gradient objective functions, and two other extensions of the algorithm are also proposed to reduce the communication and computational cost.
- In [90] a gradient descent is carried out to update the primal variable, and a gradient ascent to update the dual variable. This is the first work establishing exponential convergence without strong convexity. The convergence analysis in this publication is based on a Lyapunov function for the continuous-time algorithms; however, the authors also present the discrete counterpart of the algorithm by a discretization method based on Euler's approximation and establish the exponential convergence of the discrete-time algorithm.

As mentioned, AL-based PD methods can handle generic problems, including problems with non-smooth objective functions or problems with convex objective functions (not strongly-convex objective functions). They also have strong convergence guarantees. However, they are not applicable to problems with non-convex objective functions. For a

TABLE 1. An overview of AL-based PD methods.

Algorithm	References	Exact solution	Primal update method
MSDA	[85]–[87]	✓	Exact minimization of Lagrangian function
-	[88]	✘	Nesterov gradient method
FlexPD	[89]	✓	Multiple gradient descent steps
-	[90]	✓	Gradient descent

more detailed overview of AL-based PD methods and their complexity, we refer readers to [71]

2) ADMM

ADMM is an AL-based method, in which, the primal variables are split into M subsets ($M \geq 2$).[5] Then, instead of minimizing all the primal variables once in every iteration, as is done in ALM, the M minimization steps is done in every iteration of the algorithm (one for every subset). This way, one can benefit from the properties of different sub-problems individually. In fact, the main reason behind drawing much attention to ADMM is that when applying to recent applications mentioned above, the sub-problems are easy to solve [95]. A non-convex problem, for an instance, can be divided into some sub-problems with only one non-convex sub-problem that is easier to be solved that the first non-convex problem.

Generally, ADMM can converge to the exact solution, and its convergence rate has been proven [96], [97], [98], [99], [100]. However, many researchers have proposed inexact ADMM algorithms [93], [101] to reduce the computational cost. The inexact algorithm has attracted much attention since there is no need for solutions with extremely high accuracy in recent applications, such as those arising in artificial intelligence (AI) and big data. The inexact algorithm is usually achieved by a linear [102] or a quadratic [103] approximation of the objective function in the primal variable update.

Another extension of ADMM, driven by the applications, such as noisy color image restoration and asset allocation [104], is the case where one of the sub-problems is non-convex [104], [105], [106], [107]. These works often establish the convergence using Kurdyka–Lojasiewicz inequality [108] or similar properties.

There are also other extensions of ADMM that can handle different scenarios. These scenarios include cases with synchronous or asynchronous algorithms, fixed or time-varying networks, fixed or time-varying cost functions, smooth or non-smooth cost functions, and directed or undirected networks. In the next paragraphs of this subsection, we will discuss these extensions.

Some researchers investigated asynchronous algorithms because, in real-world implementations, some agents may fail to share their information with others on time. Subsequently, this failure will cause an immediate halt in the synchronous algorithm [109]. However, asynchronous algorithms are more robust to these failures. In [94], a modified version of a relaxed ADMM is proposed while the agents are not synchronized among themselves. The relaxed ADMM is defined as the dual of Peaceman–Rachford splitting instead of the augmented Lagrangian method. It offers more flexibility in obtaining better convergence properties, while on the other hand, it increases the complexity of the updating equations. With strongly convex and twice differentiable local objective functions, the authors of this work prove that the expectation of the error converges to a neighborhood of the optimal solution. However, no form of delay is allowed meaning agents must perform their local computations using the most updated information received from their neighbors. For non-convex objective functions, an asynchronous distributed ADMM is proposed in [110], with detailed proofs of theorems in [111] and [112], that also considers delay in the network. However, the delay needs to be bounded to ensure convergence, and the algorithm is implemented in a main-secondary mode. Another asynchronous algorithm that converges to the exact solution under the assumption of convex and possibly non-differentiable objective functions is [113]. The communications in this paper are subject to possibly unbounded delays.

ADMM algorithms have also been studied over time-varying networks since in practical applications, communication links may be lost unexpectedly. In [114], as an example, the authors developed an algorithm with a possibly stochastic gradient that works for both static and time-varying networks (but under certain assumptions, including a row-statistic weight matrix). They prove the convergence of the algorithm when the exact gradients or only noisy gradient information is known to agents. The graph is assumed to be undirected in this publication, but they use a directed graph with different edges from the original ones to represent it. This would be helpful in simplifying the definition of constraints.

There are also some algorithms designed for ADMM with time-varying cost functions. In [115], a dual-regularized ADMM is proposed, that is applicable to problems with both static and time-varying, but smooth and strongly convex cost functions. The idea of applying regularization to the dual variable comes from the fact that regularization improves the convergence properties. By doing that, the cost function of

[5] Different methods can be used to split the primal variables, such as Douglas–Rachford splitting method [93] and Peaceman-Rachford splitting method [94].

the dual problem needs to be ϵ-strongly convex instead of strongly convex ($\epsilon > 0$). They show, by numerical results, that although there is a bounded regularization error, their method can outperform inexact gradient-based methods. For time-varying cost functions that contain a non-smooth part, the publication [58] proposed an inexact algorithm. This algorithm considers a source of additive noise, representing communication noise or quantization.

Very recently, researchers have developed ADMM algorithms over directed graphs [116], [117], [118], [119] since, in practice, not all communication channels are necessarily bidirectional. In some cases, for instance, different agents have different capabilities of broadcasting messages, as a result of which, communication links between them are necessarily unidirectional. The first work in this direction is [117]. The authors of this publication proposed a directed-distributed ADMM algorithm. Under the assumptions of convex and not necessarily differentiable objective functions and column-stochastic weighted matrix, they show that their proposed algorithm converges at a rate of $O(1/t)$, with t being the iteration number. Another algorithm working over directed networks was proposed in [119]. This is the first asynchronous algorithm over directed graphs, converging at the same rate as the algorithm in [117]. However, there are some assumptions in this algorithm, such as known graph diameter[6] to all nodes.

Apart from the extensions mentioned so far, there are also other extensions of ADMM, such as accelerated [120], adaptive [121], and distributed linearized ADMM [122]. We refer interested readers in the subject to other survey papers that specifically review ADMM methods. Some examples include [123], [124], and [125], that review the classical ADMM, where $M = 2$. And, a recent paper, [95], reviews all ADMM methods, including the cases where $M > 2$.

Overall, PD methods can converge to the exact optimal solution with provable rates and can handle generic problems. For decentralized static optimization, i.e., when the objective functions are not time-varying, it is discussed in [69] that PD methods are superior to the primal methods. However, for decentralized dynamic optimization, this is not always true. On the other hand, PD methods can only be applied to problems with a simple structure of the objective function so that the primal and dual updates can be done efficiently [126].

In problems with the formulation of the sum of objective functions, that were reviewed in this section, all the objectives are prioritized equally. In practice, however, agents in a system may have various priorities for different objectives. In a fleet of self-driving cars, for instance, objectives can be travel time, stop time, or fuel consumption, and each car may prioritize them differently than others. Nevertheless, most of the existing literature formulates the MOO problem as a problem of an unprioritized sum of objective functions, which restricts the control that agents have on their priorities. In the next section, we will review publications proposing a solution to the problem formulation of the prioritized sum of objective functions with the aim of identifying the gaps in this new line of research.

[6]The diameter of a graph is the maximum distance between every pair of vertices in the graph. This distance is calculated as the number of edges in the shortest path connecting two vertices [37].

V. PROBLEM FORMULATION OF THE PRIORITIZED SUM OF OBJECTIVE FUNCTIONS

All methods reviewed so far consider the MOO problem to be unprioritized, meaning that agents cannot individually prioritize information received from other agents in the network. The MOO problem with the formulation of the prioritized sum of objective functions is defined as follows:

$$\begin{aligned} \min_{\mathbf{x}\in\mathbf{A}} \quad & \mathbf{f}(\mathbf{x}) = \sum_{i=1}^{n} w_i f_i(\mathbf{x}), \\ \text{s.t. } & \mathbf{x} \in \mathbf{X}, \\ & \mathbf{g}(\mathbf{x}) \leq 0, \\ & \mathbf{h}(\mathbf{x}) = 0, \end{aligned} \tag{4}$$

where w_i is the priority assigned to f_i.

In this section, we will conduct a review of papers proposing a solution to the problem formulation of the prioritized sum of objective functions. There are only a handful of publications proposing decentralized algorithms for solving this problem. On the other hand, there is a large body of work proposing a method to solve the problem of the prioritized sum of objective functions for centralized cases [39], [127], [128], [129]. However, these methods constrain the decision-making autonomy of agents since the central decision-maker selects the solution that best satisfies its priorities.

One of the works in this direction is [130], which solves the problem of the prioritized sum of objective functions, with local constraints ($x_1 = x_2 = \ldots = x_n$) and the priorities given as the left eigenvector of the network's weight matrix. The underlying network in this work is a directed unbalanced random network. Under the assumptions of having sub-differentiable and uniformly Lipschitz continuous objective functions, the authors showed the convergence of their algorithm to a Pareto optimal solution.

Another work that presented a protocol for the mentioned problem over an unbalanced but undirected and strongly connected network is [131]. Apart from the local constraints of having a common x_i to the network in the former work ([130]), this latter one also considers equality and inequality global constraints in the problem. The proposed protocol is based on the exact penalty method [132]. However, they replace the constant penalty parameter with a sequence. This helps guarantee the consensus and the convergence of the sequence. Here, the priorities assigned to the objective functions are again the left eigenvector of the graph's weight matrix.

Although the problem solved in [130] and [131] is formulated as the prioritized sum of objective functions, the agents do not have any control over giving different priorities to the objective functions for other agents. Indeed, the weights are given as the left eigenvector of the network's weight matrix. Therefore, the proposed algorithm does not achieve the desired autonomy of the agents, and the agents cannot individually prioritize the objectives.

This issue is addressed in [6], where an agent's priorities given to other agents' objective functions are considered in the update rule of the algorithm. That is, the priority that each agent assigns to the information received from the neighboring agents affects its decision variable. Consequently, the agents are given increased flexibility in their choices. This allows them to explore the Pareto front of the problem and find different optimal solutions along the Pareto front. In other words, agents achieve optimums that conform to their needs and preferences. The decision variables, in [6], are locally constrained to lie in a set, and this constraint set is the same for all agents. An example would be a case where agents are robots with the same range of motion. The weight matrix in the algorithm is non-doubly stochastic, the graph is connected and undirected, and the step size is assumed to be decreasing. Under these conditions, the authors in [6] proved the convergence of their proposed algorithm and provided the convergence rate. They also showed how the agents' initial choices of priorities affect the long-run behavior of the algorithm.

All of the literature reviewed in this section developed an algorithm over an unbalanced graph as they removed the need to have doubly-stochastic weight matrices. The requirement of having a sequence of doubly-stochastic matrices reflects a form of balancedness [50]. By removing it, all agents' weights assigned to a particular objective function need not sum to one. As a consequence, the applicability of these methods increases since in real-time applications, checking the balancedness of the network is not an easy task [133]. Apart from that, in some applications, communication is costly. Therefore, it is desired to avoid surplus communication needed to coordinate weights.

To the best of our knowledge, the publications reviewed in this section are the only ones with the formulation of the prioritized sum of objective functions, which are minimal numbers. This means that there is a critical gap in the literature on the subject. On the other hand, the need for these methods in the real world is growing. Many real-time applications require agents to have a high degree of flexibility in their choices.

VI. APPLICATIONS IN MULTI-ROBOT SYSTEMS

One of our focuses in this review is the applications of MOO to MRSs. MOO can be applied to many areas in MRS. Examples of these areas are the military, transportation systems, agriculture, security and surveillance, search and rescue, package delivery, household chores, health care, and elderly care. In the literature, however, only a few of these areas have been studied. We review the applications of MOO to the mentioned areas in Subsect. VI-A and the practical experiments conducted on real hardware in Subsect. VI-B. Note that applications reviewed in Subsect. VI-A are only those that have been studied in the literature. The aim is, first, to highlight the fact that most studies have only focused on the mathematical development of MOO algorithms rather than their applications. Secondly, we review the applications of MOO to MRSs to identify methods with proven applicability to real-time platforms.

A. APPLICATIONS OF MOO TO MRS

1) PATH FINDING (PATH PLANNING)

Multi-agent path finding (MAPF) is the problem of finding paths for multiple agents in a given graph such that each agent reaches its goal without colliding with other agents [134]. Recent real-world applications in robotics, AI, and operational research have led to a growing interest in MAPF. Examples of these applications could be using warehouse robots to move products rapidly and safely without human intervention [135] and using multi-robot systems to spray pesticides in agricultural fields [136].

In path-finding, agents are usually assigned to a target and have to optimize a cost function to reach their target. The cost function can be fuel consumption or completion time. In some applications, targets are predefined for each agent meaning that agents cannot be changed with each other while performing the task. However, others also include the task allocation process, which increases the agent's autonomy in managing the task at hand [137]. Some other works propose the combination of MOO algorithms with other algorithms to solve the path planning problem [138], [139].

Another issue that arises for agents when navigating to find the path is collision avoidance, which has been dealt with in some publications [137], [140]. However, centralized optimization algorithms can better treat the collision avoidance constraints than decentralized ones since in decentralized algorithms, real-time running can lead to oscillatory behavior [141].

2) TARGET TRACKING

In multi-robot target tracking, teams of robots track the position of multiple targets. The robots may be heterogeneous, meaning that they may have different sensors, and therefore, measuring various signals. Like with the path-finding application, there may also be a need to allocate tasks to robots, which may be easy or difficult depending on the number of targets and the way they are distributed.

In the literature, MOO algorithms are usually combined with other methods to track targets. In [142], the ADMM is combined with a Kalman-filter-liked update in the algorithm to track vehicles. In another work, [143], the ADMM has been used to design the interactions for multi-robot trajectory planning while the approach is based on nonlinear model predictive control (MPC).

3) FORMATION CONTROL

Formation control is the cooperation of a group of robots to complete a task, while they are required to get into and maintain a certain shape. In other words, it is controlling the relative position between agents [144]. Applications of formation control include landmine removal, search and rescue operations in dangerous environments, and surveillance [145], [146].

Among the strategies used in the field of robot formation control, MPC has been widely explored in recent years. In [144], as an instance, an ADMM-based distributed MPC is applied to the formation control of multiple underwater vehicles. In another work, [147], ADMM is again used to solve an MPC problem but with an application for the path tracking of unmanned aerial vehicles (UAVs).

From the review in this subsection, it can be concluded that most of the algorithms implemented in multi-robot systems applications are PD algorithms, especially ADMM. Overall, ADMM is a more attractive option for problems in MRS since it has a fast convergence speed and is computationally efficient [18].

B. PRACTICAL EXPERIMENTS OF MOO ON MRSS

The number of works implementing MOO problems on multi-robot hardware is minimal. This is, first, because optimization problems arising in robotics problems usually involve non-convex constraints and objective functions, while most of the existing MOO methods designed for robotics applications, such as target tracking and SLAM [18], consider the problem to be convex. In our review, the only algorithms that are designed for non-convex objective functions are extensions of ADMM with the possibility of having one non-convex sub-problem [104], [105], [106], [107] or non-convex objective functions [110], [111], [112]. As a result of this limitation, most of the designed MOO algorithms are not yet fully applicable to MRS.

Secondly, there is a limited number of real-time implementations of MOO algorithms on MRS since these algorithms often do not consider agents' communication capabilities. Although there are more papers investigating delays in communication or limited communication in both DSG and PD algorithms, they do not fully satisfy the limitations of the real world. In practice, MRSs usually operate in regions with insufficient communication infrastructure.

Similar to the previous section, all the works we found that report the results of practical experiments of implementing MOO algorithms on MRSs fall into the category of PD methods. Their number is less than a handful, showing the huge gap in the practical implementation of designed algorithms on robotic platforms. In other words, most works on the topic tend to be highly abstract and focus mainly on the mathematical proof of algorithms' convergence.

The authors of [75], which was also mentioned and explained in Subsect. IV-B, applied their proposed algorithm on a team of 8 Khepera III robots. Khepera is a mobile robot developed at the Swiss Federal Institute of Technology of Lausanne (EPFL) to be used in education and research [148]. The results in this publication indicate close agreement between the simulation and experiments.

Reference [149] presented a method of coordination for trajectory optimization of multi-robot motion planning. The method is based on a distributed nonlinear MPC. The convergence of the proposed algorithm is proved using ADMM. In their algorithm, the authors assume that each robot has an onboard computation unit to communicate with neighboring robots through a wireless network and solve a local control problem. They proposed two methods for synchronous and asynchronous information exchange between robots. They tested the asynchronous distributed nonlinear MPC to coordinate a team of three small-scale autonomous cars. The vehicles are based on a platform presented to replicate the behavior of full-scale vehicles, named the Waveshare JetRacer Pro AI5 kit [150]. The results in the paper, along with a video of the experiments provided in [151] show that the vehicles can accomplish the planning goals without collisions safely.

Another practical implementation of MOO algorithms is presented in [152]. The application is a multi-robot SLAM. Again, the authors use ADMM in the optimization process to guarantee convergence. They propose a method for outlier detection between any number of maps. The method is used in cases where multiple robots perform SLAM or one robot generates various maps. They implement their method for experiments on actual data obtained from an office environment by taking four independent sequences of RGB-D images using an Intel RealSense D435 camera. Their results show that the objects in a map are better aligned.

Again, a similar conclusion to the previous subsection can be drawn. All methods applied to MRS hardware fall into the category of PD methods. Although the number of practical implementations is low, the results show an agreement between simulation and experiments. In other words, the applicability of the PD method in practice is proved. However, the applicability of algorithms proposed to solve the problem formulation of the prioritized sum of objective functions has not been practically proved.

VII. FUTURE RESEARCH DIRECTIONS

From this mapping and review survey, we could identify important research gaps and needs that require researchers' attention as follows:

- As the review indicates, the MOO problem with the formulation of the sum of objective functions has been widely investigated in the literature. Admittedly, some subcategories have room for improvement, such as developing DSG algorithms for globally constrained MOO problems. However, the method has overall been explored by many researchers.

On the other hand, problems with the formulation of the prioritized sum of objective functions. To our knowledge, the publications reviewed in Sect. V are the only ones with the formulation of the prioritized sum of objective functions, whose number is very limited.

This means that there is a huge gap in the literature on the subject. In fact, the existing work on MOO gives agents a low level of autonomy as they consider agents' priorities to be fixed and the same for all objective functions. In other words, agents do not have any control over their priorities. Whereas, in the real world, many applications require agents to have a high degree of flexibility in their choices. Enabling them to prioritize their objectives enhances the decision-making skills of the multi-agent system, making it more robust and autonomous. However, this increased robustness and autonomy come with some disadvantages: As non-doubly stochastic weight matrices are often needed in the algorithm, the convergence analysis is more complex than problems with the sum of objective functions.

Therefore, in this new line of research that has not been explored yet, new algorithms can be developed for different scenarios, such as algorithms over time-varying networks or with directed communication. Other research directions can be considering non-convex and globally constrained cost functions.

- For both problem formulations, practical restrictions have not been usually considered in the designed algorithms. Without considering these restrictions, implementing the proposed methods would be impossible, interfering with the development of the robotics field. For instance, most of them often consider the problem to be convex, while optimization problems arising in robotics are usually non-convex. They also do not usually consider communication problems, such as delays or having limited access to communication tools.
- Another critical gap in the literature is the practical implementation of the designed algorithms. While in practice, we need reliable, robust, and safe implementation of designed algorithms, most of the studies in the literature do not implement their proposed method. Even for the problem formulation of the unprioritized sum of objective functions, the designed algorithms have not been widely implemented in practice, and most of the works on the topic focus mainly on the mathematical proof of convergence. Therefore, the applicability of the existing MOO methods and their robustness is not well assured.

  Indeed, without practical implementations or a testbed, robotic complexities cannot always be accurate the first time in simulations. In other words, we need algorithms to be implemented on practical systems to develop more accurate models and subsequently design more efficient algorithms. Therefore, implementing algorithms on real-time platforms will help in going further than simulations, getting real feedback and data in real settings, and redesigning algorithms based on them. That is to say, the real feedback will iteratively feed the development of algorithms. Examples of real feedback could be the time when resources are limited, when agents fail, when communication is noisy, or when robots interact with humans.
- Finally, in the existing practical applications, all the robots that are used are identical while in the real world, there is a need for a variety of designs to accomplish complex tasks successfully. Developing algorithms and platforms that can be applied to scenarios with different kinds of robots, such as UAVs and unmanned ground vehicles (UGVs), is, therefore a valuable research direction.

VIII. CONCLUSION

In this survey, we provided a review of MOO methods for MASs. By categorizing the existing methods based on the formulation of the problem, we discussed the main problem formulations and the distinctions between them. Our taxonomy makes it easier to identify the most recent trends in the subject and also makes it more understandable for readers to explore the subject. We also surveyed the literature on the practical applications of these methods on MRSs for the first time. This highlights the need to implement designed algorithms on real platforms while identifying reliable methods already applied to real systems. We proposed a new line of research on the subject, i.e., the problem with the formulation of the prioritized sum of objective functions.

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